

8.33

球外, $r > R$, 由高斯定理:

$$E_{\text{外}} \cdot 4\pi r^2 = \frac{Q_{\text{外}}}{\epsilon_0} = \frac{4\pi R^3}{3\epsilon_0} \rho$$

$$\Rightarrow E_{\text{外}} = \frac{\rho R^3}{3\epsilon_0 r^2}$$

球内, $r < R$, 由高斯定理:

$$E_{\text{内}} \cdot 4\pi r^2 = \frac{Q_{\text{内}}}{\epsilon_0} = \frac{4\pi r^3}{3\epsilon_0} \rho$$

$$\Rightarrow E_{\text{内}} = \frac{\rho r}{3\epsilon_0}$$

$$(1) U_{\text{外}}(r) = \int_r^\infty E_{\text{外}} dr = \int_r^\infty \frac{\rho R^3}{3\epsilon_0 r^2} dr = \frac{\rho R^3}{3\epsilon_0 r}$$

$$(2) r=R \text{ 时, } U_{\text{表}} = U_{\text{外}}(R) = \frac{\rho R^2}{3\epsilon_0}$$

$$(3) \text{ 当 } r < R \text{ 时, } U_{\text{内}} = \int_r^R E_{\text{内}} dr + U_{\text{表}} = \int_r^R \frac{\rho r}{3\epsilon_0} dr + \frac{\rho R^2}{3\epsilon_0} = \frac{\rho}{6\epsilon_0} (3R^2 - r^2)$$

8.34

(3) 设 $A(a, 0)$, 在平面 $x=a$, P 在该平面设 $P(a, r)$ dx 离 P 距离: $\sqrt{(a-x)^2 + r^2}$.

$$U = \int_{-\infty}^{\infty} \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{\sqrt{(a-x)^2 + r^2}} = \frac{\lambda}{4\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{dx}{\sqrt{(a-x)^2 + r^2}} = \frac{\lambda}{4\pi\epsilon_0} \ln \left| \frac{2a + \sqrt{4a^2 + r^2}}{r} \right|$$

$$E_{AP} = - \frac{\partial U}{\partial r} = \frac{\lambda}{4\pi\epsilon_0 r \sqrt{4a^2 + r^2}}$$

8.34 $\lambda = \frac{q}{2a}$, 设无穷远处为电势零点, dx 处电荷 $dq = \lambda dx$.在距离为 R 的点产生的微元为 $dU = \frac{dq}{4\pi\epsilon_0 R} = \frac{\lambda dx}{4\pi\epsilon_0 R}$.(1) P 在线段垂直平分面上时, 以 O 为原点, 线段所在直线为 x 轴. 设 $P(0, r)$, 线段为 $[-a, a]$.

$$U = \int_{-a}^a \frac{\lambda dx}{4\pi\epsilon_0 \sqrt{x^2 + r^2}} = \frac{\lambda \cdot 2}{4\pi\epsilon_0} \int_0^a \frac{dx}{\sqrt{x^2 + r^2}} = \frac{\lambda}{2\pi\epsilon_0} \ln \left(\frac{a + \sqrt{a^2 + r^2}}{r} \right)$$

$$E_{op} = - \frac{\partial U}{\partial r} = - \frac{\lambda}{2\pi\epsilon_0} \left(- \frac{1}{a + \sqrt{a^2 + r^2}} \cdot \frac{r}{\sqrt{a^2 + r^2}} - \frac{1}{r} \right) \left(\lambda a = \frac{q}{2} \right)$$

$$= \frac{q}{4\pi\epsilon_0 r \sqrt{a^2 + r^2}} \cdot \left(1 = \frac{\lambda a}{2\pi\epsilon_0 r \sqrt{a^2 + r^2}} \right)$$

$$(2) P(r, 0), r > a, U = \int_{-a}^a \frac{\lambda dx}{4\pi\epsilon_0 (r-x)} = \frac{\lambda}{4\pi\epsilon_0} \ln \left(\frac{r+a}{r-a} \right)$$

$$\text{代 } \lambda = \frac{q}{2a} \Rightarrow U = \frac{q}{4\pi\epsilon_0 a} \ln \left(\frac{r+a}{r-a} \right)$$

$$E_{op} = - \frac{\partial U}{\partial r} = \frac{q}{4\pi\epsilon_0 a (r^2 - a^2)}$$

8.40 机械能守恒. 设 $e = 1.6 \times 10^{-19} \text{C}$

$$U = U_+ - U_- = 300 \text{V}$$

动能定理, $W = (-e)(U - U_+) = 4.8 \times 10^{-17} \text{J}$

或 $E_k = 300 \text{eV}$

(2) 设两圆柱极板单位长度上所带电荷量线密度为 λ .

由对称性知, 两板间距离轴线为 r 处的 E 仅有径向分量.
取半径 r 、长度 L 的同轴圆柱面为高斯面.

由高斯定理: $\Phi_E = E \cdot 2\pi r L = \frac{Q_{\text{内}}}{\epsilon_0} = \frac{\lambda L}{\epsilon_0}$

$$\Rightarrow E(r) = \frac{\lambda}{2\pi\epsilon_0 r}$$

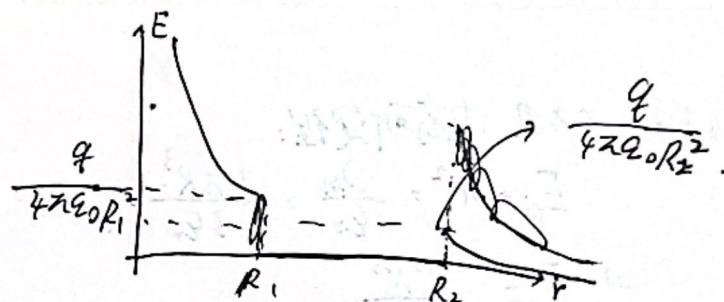
而 $U_+ - U_- = \int_{R_1}^{R_2} E(r) dr = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{R_2}{R_1}\right)$

$$\Rightarrow \frac{\lambda}{2\pi\epsilon_0} = \frac{U_+ - U_-}{\ln\left(\frac{R_2}{R_1}\right)}$$

代入 $E(r) = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{1}{r} \frac{U_+ - U_-}{\ln\left(\frac{R_2}{R_1}\right)}$

□

8.41



由球对称性, 取半径为 r 的同心高斯球面. 由高斯定理.

(a) 球壳内, $0 < r < R_1$:

$$E \cdot 4\pi r^2 = \frac{q}{\epsilon_0} \Rightarrow E = \frac{q}{4\pi\epsilon_0 r^2}$$

(b) 导体球壳内, $R_1 \leq r \leq R_2$, $E = 0$ (静电平衡)

(c) 球壳外, $r > R_2$:

$$q + (-q) + q = q$$

$$E \cdot 4\pi r^2 = \frac{q}{\epsilon_0} \Rightarrow E = \frac{q}{4\pi\epsilon_0 r^2}$$

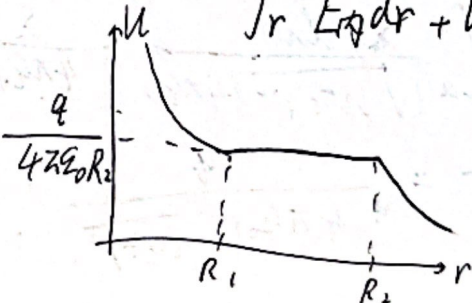
电势: $U(r) = \int_r^\infty E dr$ (以无穷远为电势零点).

① 由外向内积. ① $r > R_2$: $U(r) = \int_r^\infty \frac{q}{4\pi\epsilon_0 r^2} dr = \frac{q}{4\pi\epsilon_0 r}$

$r = R_2$ 时, $U_{R_2} = \frac{q}{4\pi\epsilon_0 R_2}$

② $R_1 \leq r \leq R_2$; $E = 0$, $U(r) = U(R_2) = \frac{q}{4\pi\epsilon_0 R_2}$

③ $r < R_1$, $U(r) = \int_r^{R_1} E dr + U(R_1) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r} + \frac{1}{R_2} - \frac{1}{R_1} \right)$



8.56. $C = \frac{\epsilon_r \epsilon_0 S}{d}$

(a) $C_1 = \frac{\epsilon_r \epsilon_0 \cdot \frac{S}{2}}{d} = \frac{\epsilon_r \epsilon_0 S}{2d}$
 并联 $C_2 = \frac{\epsilon_r \epsilon_0 S}{2d} \Rightarrow C_a = C_1 + C_2 = \frac{\epsilon_0 S}{2d} (\epsilon_r + \epsilon_r)$

(b) $C_1 = \frac{\epsilon_r \epsilon_0 S}{\frac{d}{2}} = \frac{2\epsilon_r \epsilon_0 S}{d}$
 串联 $C_2 = \frac{2\epsilon_r \epsilon_0 S}{d}$
 $\frac{1}{C_b} = \frac{1}{C_1} + \frac{1}{C_2} \Rightarrow C_b = \frac{2\epsilon_0 S}{d} \frac{\epsilon_r}{\epsilon_r + \epsilon_r}$

(c) 混联 $C_{左} = \frac{\epsilon_r \epsilon_0 S}{2d}$ $C_{右} = \frac{1}{\frac{1}{C_2} + \frac{1}{C_3}} = \frac{\epsilon_0 S}{d} \frac{\epsilon_r \epsilon_r}{\epsilon_r + \epsilon_r}$

2) $C = C_{左} + C_{右} = \frac{\epsilon_0 S}{2d} (\epsilon_r + \frac{\epsilon_r \epsilon_r}{\epsilon_r + \epsilon_r})$

补充2: $C_0 = \frac{\epsilon_0 S}{d}$, $Q_0 = C_0 U = \frac{\epsilon_0 S U}{d}$

断开后, $Q = Q_0$ 不变.

1) 等效为两个并联电容器, $C_1 = \frac{\epsilon_0 \cdot \frac{S}{2}}{d} = \frac{\epsilon_0 S}{2d}$

$C_2 = \frac{\epsilon_r \epsilon_0 \cdot \frac{S}{2}}{d} = \frac{\epsilon_r \epsilon_0 S}{2d}$

$C' = C_1 + C_2 = \frac{1 + \epsilon_r}{2} C_0$, $W_e = \frac{Q_0^2}{2C'} = \frac{\epsilon_0 S U^2}{d(1 + \epsilon_r)} (= \frac{C_0 U^2}{1 + \epsilon_r})$

2) 并联, 具有相同电压 $U' = \frac{Q_0}{C'} = \frac{2U}{1 + \epsilon_r} \Rightarrow E' = \frac{U'}{d} = \frac{2U}{d(1 + \epsilon_r)}$

8.58 电介质层, $E_1 = \frac{D}{\epsilon_r \epsilon_0}$, 真空层, $E_2 = \frac{D}{\epsilon_0}$

$\Rightarrow E_2 = \epsilon_r E_1$

1) $\oint U = E_1 t + E_2 (d - t)$

$\Rightarrow U = E_1 [t + \epsilon_r (d - t)]$

$\Rightarrow E_1 = \frac{U}{t + \epsilon_r (d - t)}$

$D = \epsilon_0 \epsilon_r E_1 = \frac{\epsilon_0 U}{\frac{t}{\epsilon_r} + d - t}$

2) 极板上自由电荷面密度 σ 数值上
与电位移 D 大小相等.

$\sigma = D = \frac{\epsilon_r \epsilon_0 U}{t + \epsilon_r (d - t)}$

$Q = \sigma S = \frac{\epsilon_r \epsilon_0 S U}{t + \epsilon_r (d - t)}$

3) $C = \frac{Q}{U} = \frac{\epsilon_r \epsilon_0 S}{t + \epsilon_r (d - t)}$

由高斯定理, $\sigma = D$:

$\sigma_1 = D_1 = \epsilon_0 E' = \frac{2\epsilon_0 U}{d(1 + \epsilon_r)}$

$\sigma_2 = D_2 = \epsilon_r \epsilon_0 E' = \frac{2\epsilon_r \epsilon_0 U}{d(1 + \epsilon_r)}$

5.60

ii) 作半径为 r ($R_1 < r < R_2$) 长为 L 的同轴高斯圆柱面。

由高斯定理及轴对称: $D \cdot 2\pi r L = \lambda L \Rightarrow D = \frac{\lambda}{2\pi r}$.
(D 仅有径向分量)

D 仅与到轴线的距离有关, 在分界面 R_i 处连续。

$$\text{由 } E = \frac{D}{\epsilon_r \epsilon_0}$$

$$\text{① } R_1 < r < R_2 \text{ 时, } E_1 = \frac{\lambda}{2\pi \epsilon_{r1} \epsilon_0 r}$$

$$\text{② } R_2 < r < R_3 \text{ 时, } E_2 = \frac{\lambda}{2\pi \epsilon_{r2} \epsilon_0 r}$$

$$\text{iii) } U = \int_{R_1}^{R_3} E \cdot dr = \int_{R_1}^{R_2} E_1 dr + \int_{R_2}^{R_3} E_2 dr$$

$$\text{代入得 } U = \frac{\lambda}{2\pi \epsilon_0} \left[\frac{1}{\epsilon_{r1}} \ln\left(\frac{R_2}{R_1}\right) + \frac{1}{\epsilon_{r2}} \ln\left(\frac{R_3}{R_2}\right) \right]$$

iv) 设长为 L 电容器总电荷量为 $Q = \lambda L$ 。

$$\text{由 } C = \frac{Q}{U}, C_L = \frac{C}{L} = \frac{Q}{UL} = \frac{\lambda}{U}$$

$$\Rightarrow C_L = \frac{2\pi \lambda}{\frac{\lambda}{2\pi \epsilon_0} \left[\frac{1}{\epsilon_{r1}} \ln\left(\frac{R_2}{R_1}\right) + \frac{1}{\epsilon_{r2}} \ln\left(\frac{R_3}{R_2}\right) \right]}$$

补充题1: $d^2 \ll S$, 故忽略边缘效应, 视各层内部电场视为匀强电场。

$$A \text{ 与 } C \text{ 之间: } \Delta U_{AC} = U - U_C$$

$$E_1 = \frac{U - U_C}{\frac{d}{2}} = \frac{2(U - U_C)}{d}$$

$$\text{规定向上为正: } \sigma = -\epsilon_0 E_1 = -\frac{2\epsilon_0(U - U_C)}{d}$$

$$C \text{ 与 } B \text{ 之间: } \Delta U = U_C - 0$$

$$E_2 = \frac{U_C - 0}{\frac{d}{2}} = \frac{2U_C}{d}$$

C 下表面自由电荷面密度:

$$\sigma' = \epsilon_0 E_2 = \frac{2\epsilon_0 U_C}{d}$$

$$\text{由电荷守恒: } Q_C = (\sigma + \sigma') S = q$$

$$\Rightarrow \left(-\frac{2\epsilon_0(U - U_C)}{d} + \frac{2\epsilon_0 U_C}{d} \right) \cdot S = q$$

$$\Rightarrow \text{由 } C_0 = \frac{\epsilon_0 S}{d}, \Rightarrow 2C_0 U_C - 2C_0 U = q$$

$$\Rightarrow U_C = \frac{U}{2} + \frac{qd}{4\epsilon_0 S}$$